

Figure replication showcase: Mehregan et al.

Enhancing Adaptive Deep Brain Stimulation via Efficient Reinforcement Learning

This is a concise visual summary of the initial qualitative replications of the paper's plant and training panels (Fig. 1b through Fig. 4b). Each section presents the original paper panel alongside the reproduction, with notes on the physical and numerical properties verified.

I built this from the paper text and figures (no released training or environment source). The shared dynamics use the [Kumaravelu et al. \(2016\)](#) cortex-basal ganglia-thalamus model. Full run commands, manifests, and internal gates live in [figures/mehregan/replications.md](#).

Overview (July 14, 2026): Fig. 1b, 2a, 2b, 4a, and 4b pass qualitative gates. Post-training efficacy (Fig. 5) and quantization panels (Fig. 6) are still in progress.

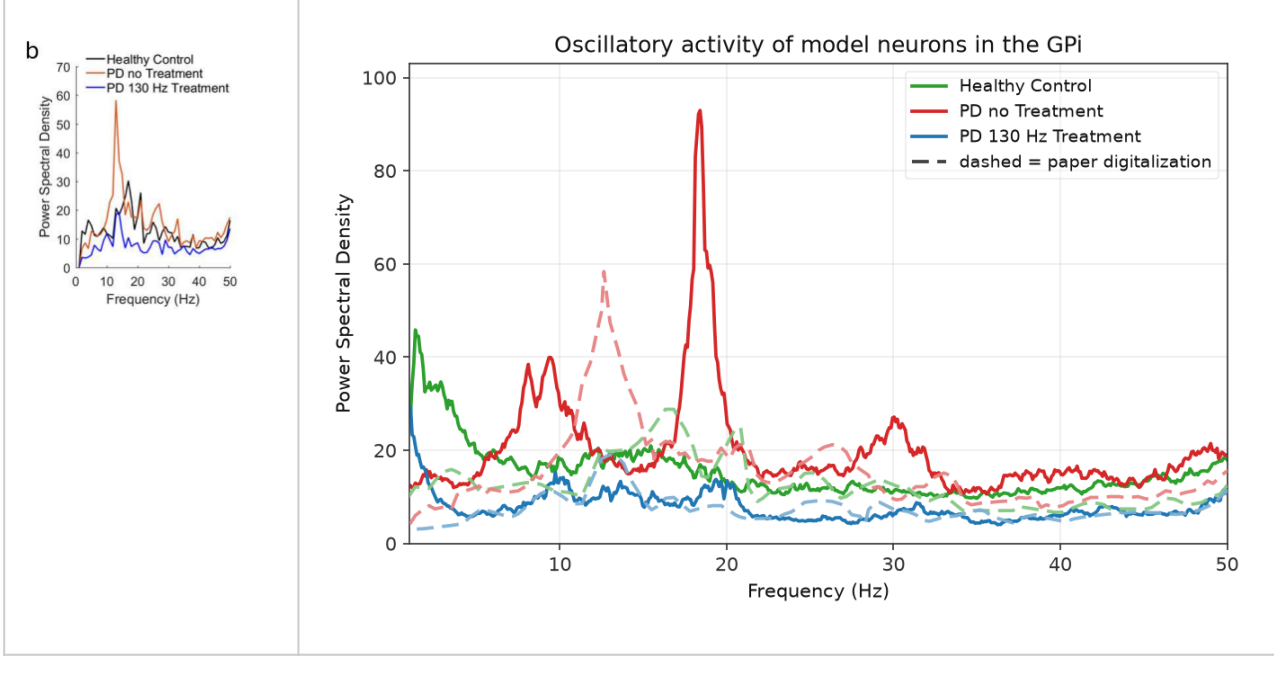
At a glance

Panel	What it shows	Result
Fig. 1b	GPI power spectrum, healthy, PD, PD + 130 Hz cDBS	Pass
Fig. 2a	GPI beta power over time, PD vs PD + cDBS	Pass
Fig. 2b	Error Index over time, PD vs PD + cDBS	Pass
Fig. 4a	Training curve, beta power vs RL step (45 Hz init)	Pass
Fig. 4b	Training curves, reward and episode-mean beta vs episode	Pass

Fig. 1b: GPI power spectral density

Paper claim: Parkinson's disease elevates GPI beta-band power relative to healthy controls; **130 Hz** continuous STN-DBS suppresses that elevation.

What I checked: Correct **ordering** across the three conditions (PD > healthy on beta power; cDBS < untreated PD).

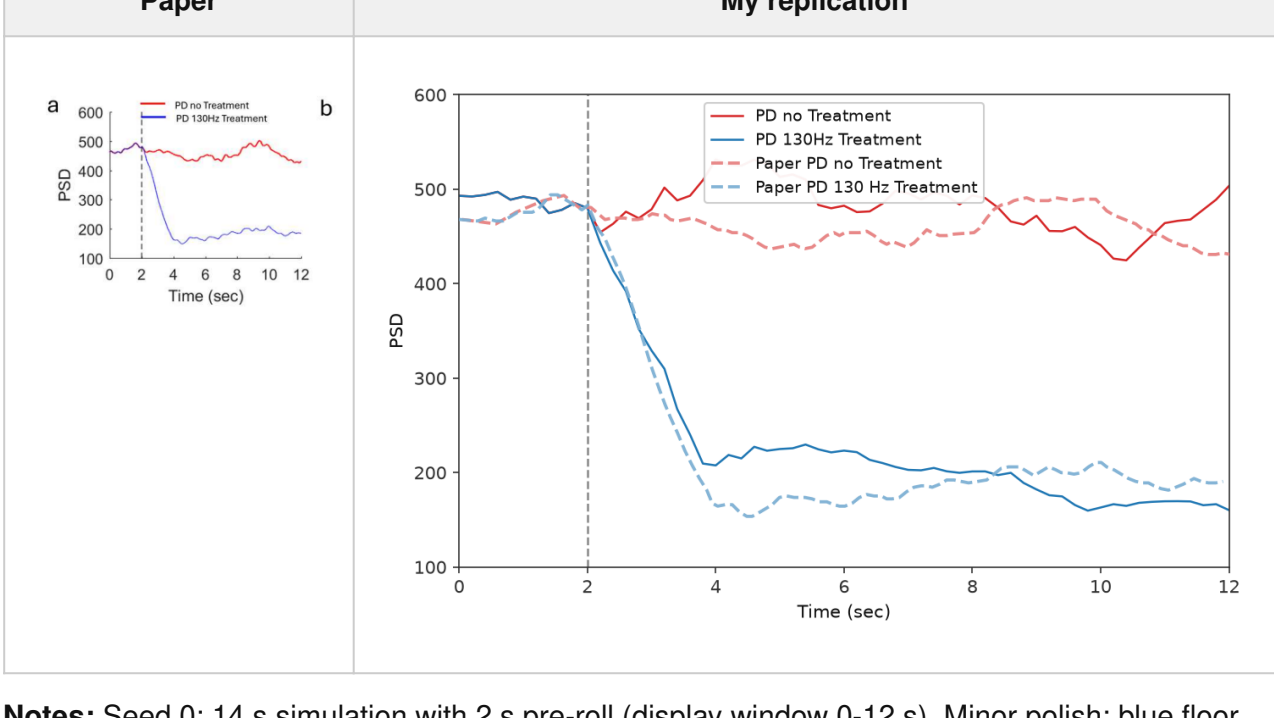


Notes: Mean PSD over seeds 0-9, 10 s segment, native Python plant port of the Kumaravelu model.

Fig. 2a: GPI beta power time series

Paper claim: After cDBS turns on at **2 s**, beta power in the treated (blue) trace **falls** and stays below the untreated PD (red) trace. Both traces share the same pre-stimulus baseline.

What I checked: Shared 0-2 s baseline, blue below red after onset, dense trailing-window protocol aligned with the paper's 12 s display window.

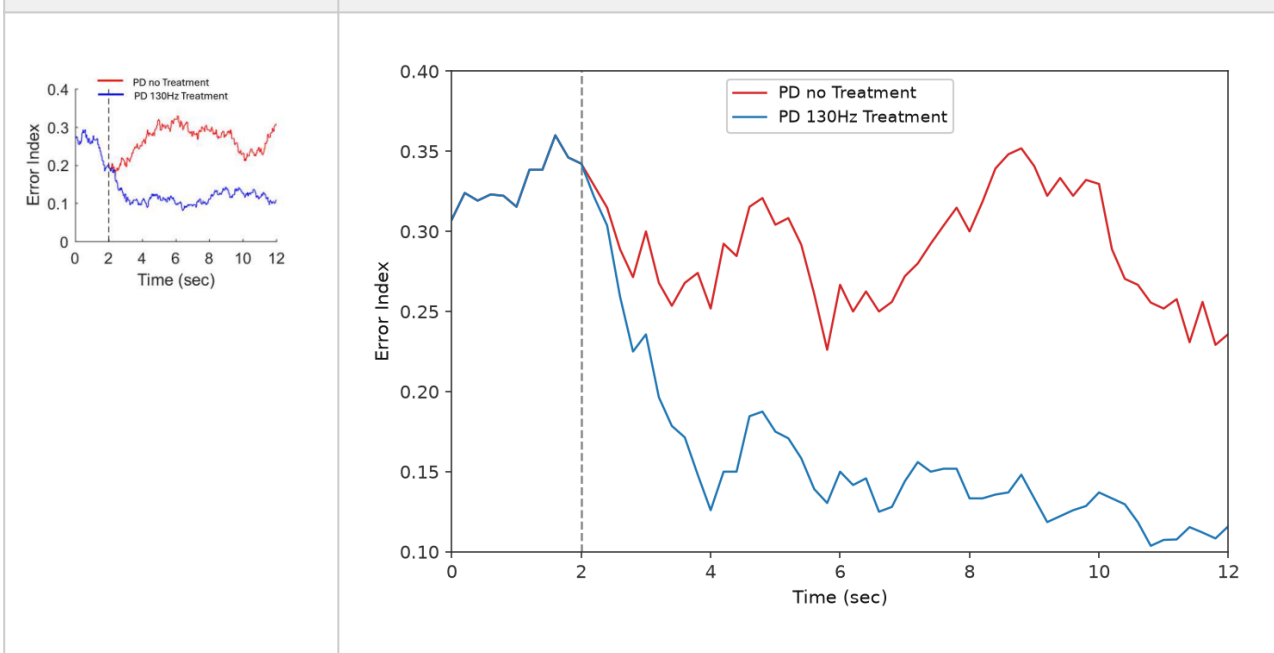


Notes: Seed 0; 14 s simulation with 2 s pre-roll (display window 0-12 s). Minor polish: blue floor slightly below the paper at late times.

Fig. 2b: Error Index time series

Paper claim: Same timing as Fig. 2a, but the biomarker is the **Error Index** (windowed thalamic spike-timing metric). Treated PD (blue) sits **below** untreated PD (red) after cDBS onset.

What I checked: Ordering after t=2 s and a shared 0–2 s baseline.

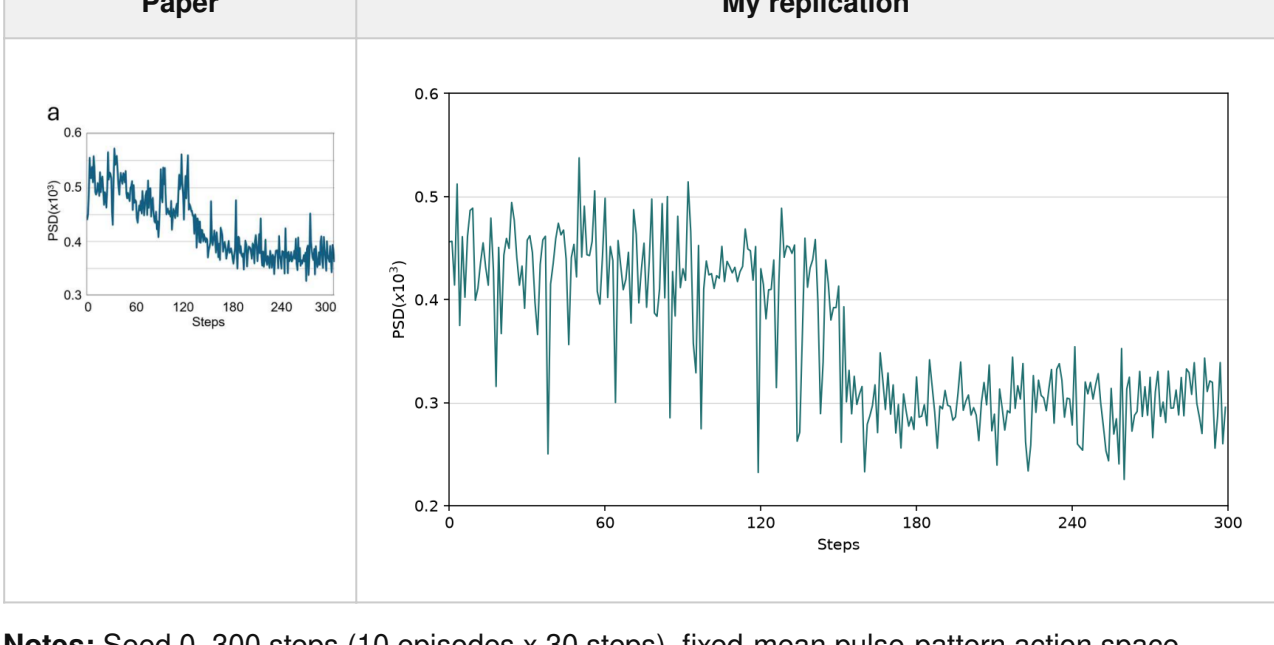


Notes: The Kumaravelu reference uses constant thalamic bias current instead of the So et al. (2012) SMC pulse drive the Error Index metric assumes. For this panel I restored **So-style SMC pulses into thalamus** (documented hybrid convention in [plant.md](#)). Seed 0.

Fig. 4a: Training beta power vs step

Paper claim: During **45 Hz** DDPG training, per-step GPI beta power is noisy early, then **drops sharply** around steps 130-150 and settles lower.

What I checked: Qualitative training **shape**, high early variance, mid-run drop, lower late plateau, on the same PSD scale as the paper panel.

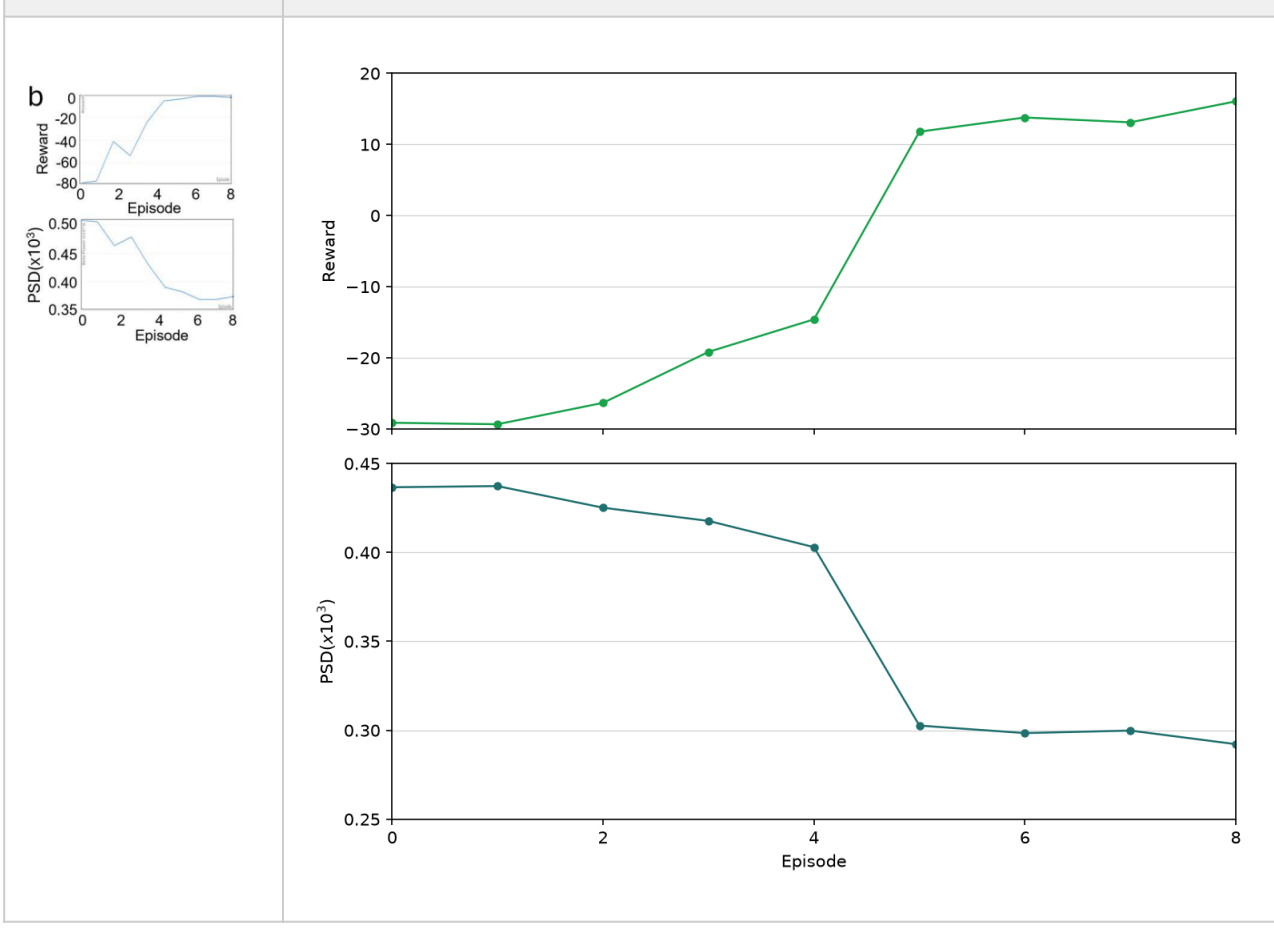


Notes: Seed 0, 300 steps (10 episodes x 30 steps), fixed-mean pulse-pattern action space, softmax exploration with one-hot critic input. Late mean sits a bit below the paper band; I treat that as numeric polish, not a gate failure.

Fig. 4b: Training reward and episode-mean beta

Paper claim: Over **9 episodes** (indices 0-8), **total reward rises** toward zero while **episode-mean beta power falls**, the inverse relationship expected from the reward definition.

What I checked: Paired with the Fig. 4a training run: reward trend up by roughly episodes 3-5, episode-mean PSD trend down. Same stacked layout as the paper (reward top, PSD bottom).



Notes: Same seed-0 training cache as Fig. 4a v4. Separate reward/PSD PNGs remain in the repo for debugging (training_reward_v14.png, training_psd_v14.png). The paper does not report the training RNG seed; different seeds change wiggles and levels. I compare **trends**, not pointwise values.

How this was produced

- Plant.** Kumaravelu et al. (2016) CBGT model (MATLAB reference + native Python port); GPI beta biomarker per Mehregan Eq. (1), 13-35 Hz.
- Environment.** Gymnasium-style 2 s steps, reward Eq. (8), fixed-mean pulse-pattern alphabet at 45 Hz for section IV.A.1 training panels.
- Controller.** DDPG actor-critic architecture per Fig. 3; hyperparameters from section IV.A.1.
- Validation.** Qualitative gates (ordering, onset timing, training shape) documented in [figures/mehregan/replications.md](#) and [replication-fidelity.md](#).

Reproduce any panel:

```
git clone https://github.com/wilzen3476/rl-adaptive-dbs.git
cd rl-adaptive-dbs

uv run python scripts/figures/papers/mehregan/1b/plot.py # Fig. 1b
uv run python scripts/figures/papers/mehregan/2a/plot.py # Fig. 2a
uv run --group figures python scripts/figures/papers/mehregan/2b/plot.py # Fig. 2b
uv run python scripts/figures/papers/mehregan/4a/plot.py # Fig. 4a (long run)
uv run python scripts/figures/papers/mehregan/4b/plot.py --plot-only # Fig. 4b from 4a cache
```

Environment: All runs were executed and verified locally using uv. Standard setup procedures are provided in scripts/setup.sh.

What's next

Panel	Topic	Status
Fig. 5a / 5b	Post-training efficacy (45 Hz / 30 Hz)	Open
Fig. 6a / 6b	PTQ / QAT comparison	Open

Key areas for ongoing investigation include the thalamic drive convention in Fig. 2b and training-seed sensitivity across Figs. 4a–4b.